ISHA PATRO

ip2529@nyu.edu | +1 (347) 241-6603 | https://ishapatro.in/ | New York, 11213

EDUCATION

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING Brooklyn, NY
Master of Science in **Financial Engineering** Expected 05/27
GPA: 3.6/4.0

VELLORE INSTITUTE OF TECHNOLOGY, AP Andhra Pradesh, India
Bachelor in **Computer Science** 07/2
GPA: 3.5/4.0

PROGRAMMING & TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C++, R, VBA
Web Development: React, RESTful APIs, AG Grid, Streamlit, Dash, Tablue
Data Science: Statistical Modeling, Machine Learning, Deep Learning, LLMs, RAG, AI Agents, Causal Inference
Databases & Cloud: SQL, Neo4j, q/kdb+, AWS, MongoDB, Microsoft Excel, Bloomberg Terminal
Quantitative Finance: Financial Modeling, Portfolio Management, Fixed Income, Time Series Forecasting, Prediction Markets

EXPERIENCE

J.P MORGAN CHASE & CO. 01/21 - 08/25
Software Developer Engineer - II, MBS Division - Fixed Income Pricing Team (Banglore, India)
- Led the development and post-launch ownership of Trade Activity Monitoring (TAM) as Epic Owner, collaborating with stakeholders to achieve a 50% latency reduction and significantly improve trading floor execution efficiency.
- Pioneered and architected a comprehensive Mortgage Matrix modernization initiative driving 12% latency improvement supporting $5B+ annual trading volume.
- Designed and implemented Trade Reporting and Compliance Engine (TRACE) in an agile environment collaborating with cross-functional teams to boost data processing efficiency by 30% and streamline regulatory reporting workflows.

RESEARCH & OPEN SOURCE CONTRIBUTION

Quantitative Finance Corpus Architecture 07/26
- Collaborated with finance professionals to curate a rigorous, error-annotated corpus of complex fixed-income and derivatives problems, guiding AI agents to validate calculations against Excel outputs, improving mathematical precision and reducing hallucinations in quantitative finance AI systems.

FinOKF - Clearbox Financial Research and Caching System 06/25 - Present
Under review at Journal of Association for the Advancement of Artificial Intelligence
- Developing a clear-box financial RAG and enterprise caching platform leveraging open-weight models to transform SEC EDGAR filings and portfolio context into structured knowledge graphs for low-latency, hallucination-resistant equity research.

Narrative Risk Allocation for Commodity Portfolios 01/26 - Present
- Engineering an end-to-end differentiable deep-learning portfolio allocator that embeds 125K+ financial news articles into volatility-spillover signals across 9 commodity futures, achieving a 1.73 Sharpe ratio over a 16-year walk-forward backtest.

Implied Prepayment Surfaces via Finance-Informed Neural Networks in Agency MBS Markets 11/25 - 03/26
Under review at Journal of Computational Finance.
- Built a finance-informed neural network that inverts agency TBA prices into a market-implied prepayment (CPR) surface under a Hull–White no-arbitrage constraint, showing the implied–realized divergence is a predictor of forward TBA returns

qutePandas (Open Source Contribution) 06/25 - Present
- Published a high-performance Pandas-compatible execution layer backed by kdb+/q, achieving order-of-magnitude speedups(10x–100x+) on large-scale analytical workloads while preserving Python-first research ergonomics.

Unlocking Financial Wisdom: Building Knowledge Graphs from Investopedia 06-24 - 08/24
Published in International Journal of Research and Analytical Reviews (IJRAR), Sep 2024
- Built a financial knowledge graph from 6,000+ Investopedia terms using data extraction and relationship mapping, enabling semantic visualization in Obsidian; presented at PitchWell Season 3 Finals at J.P. Morgan

HONORS / AWARDS

- Awarded the prestigious **Numerix Women in Finance Scholarship**, a **$20,000 honor**, 2025
- Two-time 1st Place Winner, Engineering Clinics Expo, VIT (**$1,200**) - across all engineering disciplines, 2017-201

ACADEMIC PROJECTS

2WIN 06/26
- Designed and backtested quantitative prediction-market strategies using tick-level order-book data, dynamic risk-based sizing, automated execution logic, and portfolio PnL attribution across real-time Polymarket markets.

VORTEX 05/26
- Architected a full-stack quantitative risk analytics platform: Gaussian HMM regime detection, 5-method VaR backtesting, XGBoost volatility forecasting, and a LangGraph multi-agent LLM system.

CryptoHedgeLab 03/26

- Engineered a high-performance crypto trading and backtesting system in C++ with kdb+/q, achieving microsecond-level execution, real-time hedging, and tick-level data storage for deterministic strategy evaluation.

Cross-Currency Dynamics in Cryptocurrencies under Stablecoin Regulation 01/26 - 03/26

- Analyzed cross-currency basis and liquidity fragmentation in crypto markets using high-frequency data across exchanges, evaluating stablecoin-driven pricing inefficiencies and regulatory impact (GENIUS Act)

Monte Carlo Pricing of European Barrier Options 10/25

- Developed a Monte Carlo engine under risk-neutral GBM to price path-dependent barrier options (Down-and-Out, Knock-In/Out), incorporating barrier monitoring and sensitivity analysis.

Advanced NASDAQ Market Prediction System with Deep Learning Optimization 08/25

- Engineered a sophisticated LSTM neural network system for high-frequency NASDAQ market prediction, achieving 26.7% improvement in forecasting accuracy through advanced hyperparameter optimization.

Minted Wealth 03/25

- Developed a comprehensive financial intelligence dashboard integrating live market data, portfolio performance analytics, and net worth insights for smarter investment decisions.